QuantHarness: Price-Driven Multi-Agent LLMs for High-Frequency Trading

Fei Xiong, Xiang Zhang, Aosong Feng, Siqi Sun, Chenyu You

Paper review by Your Name

October 22, 2026

Motivation and Research Question

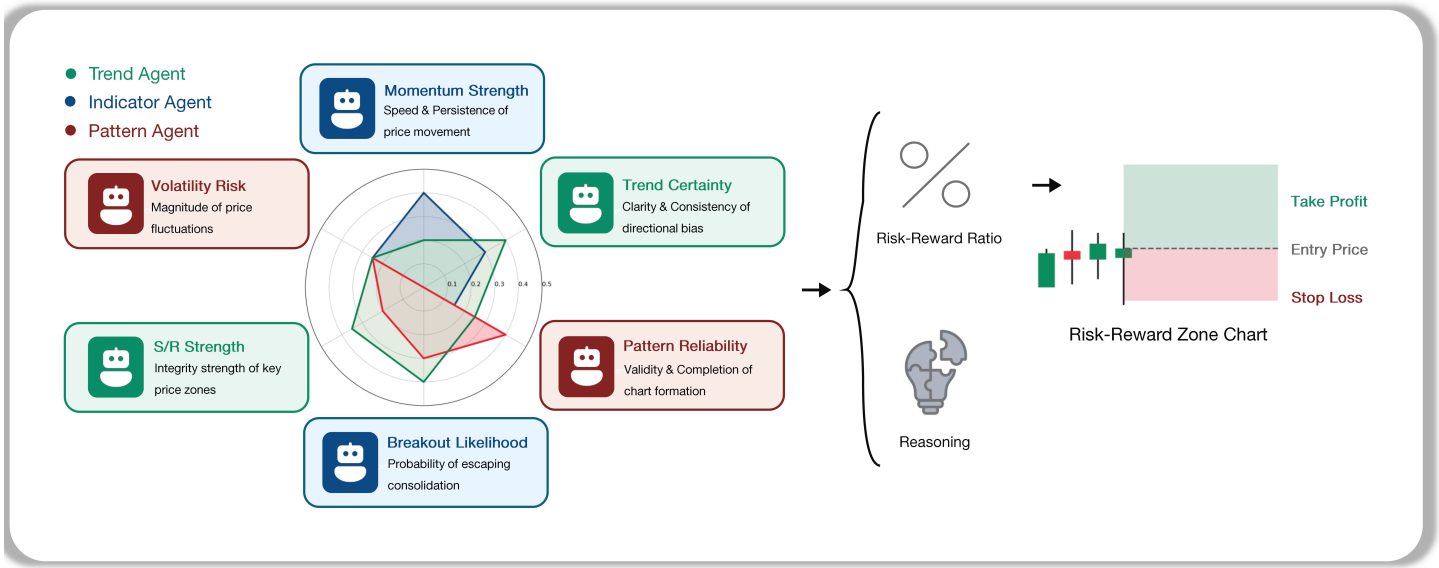
Most LLM trading frameworks use text data for making decisions.
QuantHarness reads only prices.

	Most LLM trading frameworks	**QuantHarness**
Input	News, social media, earnings reports	Price candles only (OHLC)
Horizon	Long-term	The next 3 candles of 1h or 4h bars
Signal	Lags the price, and is noisy, unstructured and hard to verify	Available as soon as a candle closes, and structured

Research question: can LLM agents that only see price data predict short-term direction better than simple baselines?

QuantHarness Architecture

- QuantHarness is a collaborative framework for low-latency market decision-making which operates solely on price-derived market signals.
- In the code, the system splits trading into four specialized agents: IndicatorAgent, PatternAgent, TrendAgent, and DecisionAgent.

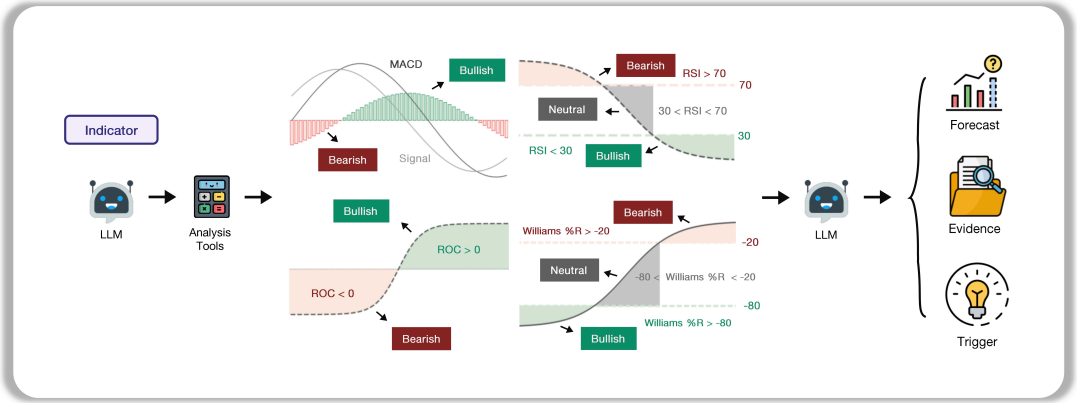


The paper's RiskAgent (Fig. 2) turns the three reports into a risk–reward setup; in the code, the DecisionAgent does this

Indicator and Pattern

IndicatorAgent

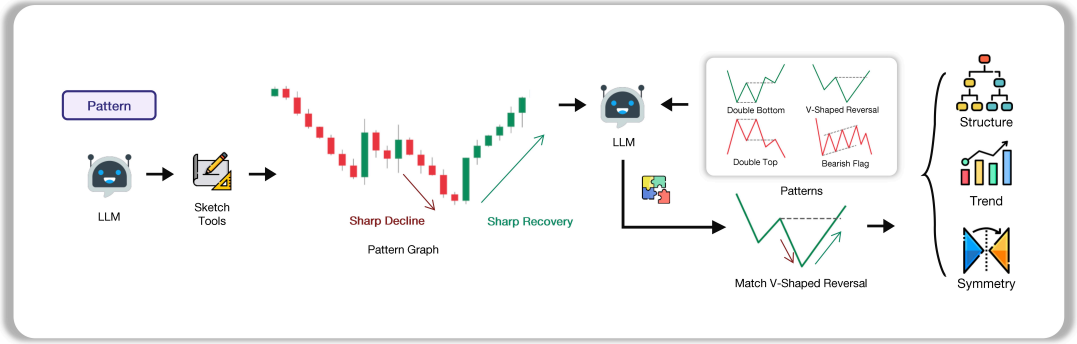
- Computes RSI, MACD, RoC, Stochastic, Williams %R on the last 45 candles
- Writes a short momentum report



Thresholds turn indicators into signals, e.g. RSI above 70 is bearish, below 30 bullish (paper, Fig. 1)

PatternAgent

- Gets a candlestick chart of the same candles
- Picks the closest of 16 named patterns

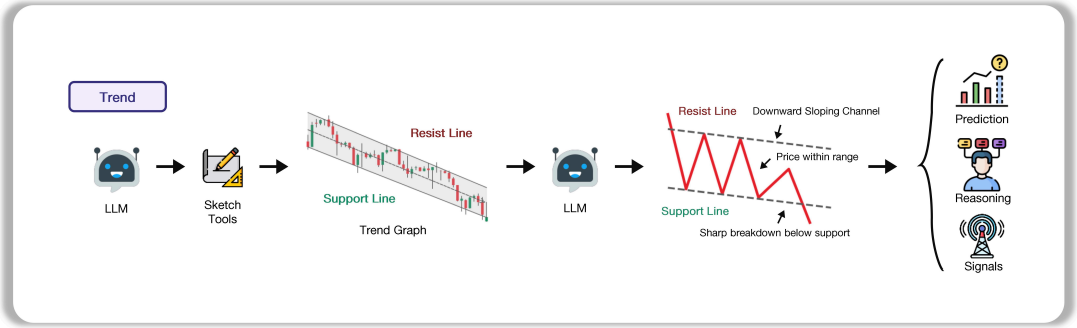


A sketch of the price moves is matched against a library of patterns, here a V-shaped reversal (paper, Fig. 1)

Trend and Decision

TrendAgent

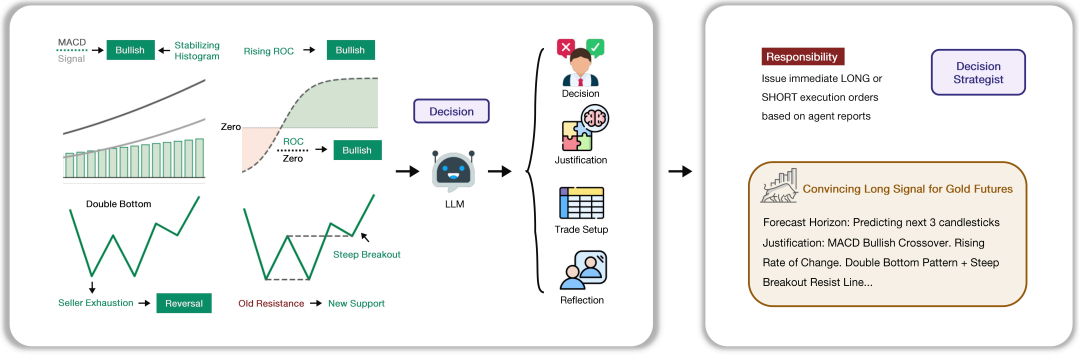
- Gets the candles with fitted support and resistance lines
- Calls the short-term trend upward, downward or sideways



A downward channel that breaks below support is read as bearish (paper, Fig. 1)

DecisionAgent

- Combines the three reports
- Outputs LONG or SHORT with a justification and a risk–reward ratio



Bullish evidence from the three reports becomes a LONG order with its justification (paper, Fig. 3)

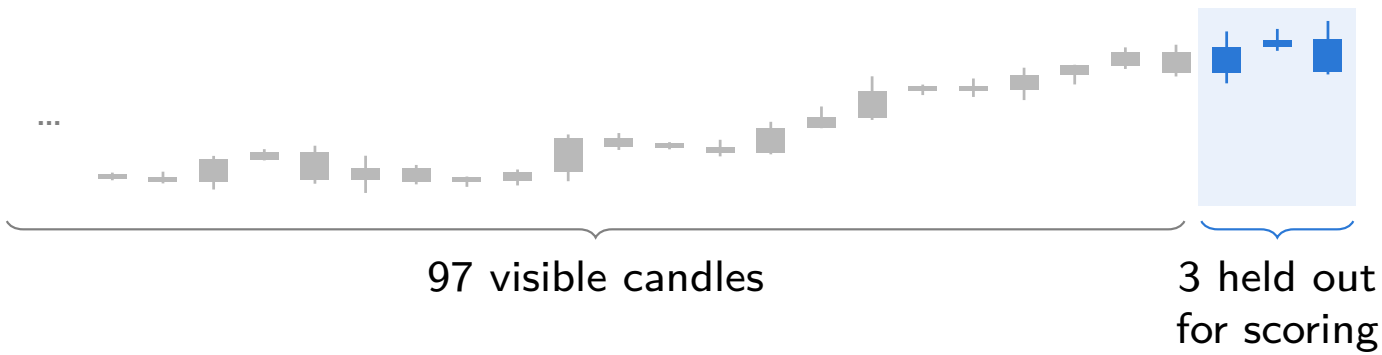
Note: the paper describes a separate RiskAgent for stop-loss and take-profit; in the code it is just one line of the DecisionAgent prompt asking for the risk–reward ratio.

Data

Asset	Type	4h	1h
BTC	Crypto	2023–25	2025
CL	Crude oil futures	2022–25	2024–25
ES, NQ	Index futures	2022–25	2024–25
DJI, SPX	Index (spot)	2015–25	2022–25
QQQ	ETF	2015–25	2022–25
VIX	Volatility index	2020–25	–
DAX	Index futures	–	2024–25

From the paper's Tables 3–4; the repo's files match exactly

- 1h and 4h OHLC bars from YFinance and TradingView
- 5,000 bars per asset and timeframe
- 100 random segments of 100 candles each
- Fewer trading hours mean a longer history: up to 10 years at 4h



Baselines

Every method sees the 97 visible candles of a segment and calls LONG or SHORT for the next 3

Method	Input	Decision
Random	–	LONG or SHORT at random; risk–reward ratio drawn from $[1.2, 1.8]$
Linear regression	Last 40 closes	Slope of the fitted line: positive means LONG, otherwise SHORT
XGBoost	RSI, MACD and SMA from TA-Lib	Trained on 50 random files, tested on the other 50; majority vote, HOLD calls dropped
QuantHarness	Last 40–50 candles	Indicator, Pattern, Trend and Decision LLM agents

Evaluation Metrics

t is the last visible candle; C, H, L are close, high and low; $d = +1$ for LONG and -1 for SHORT

Accuracy: share of the next 3 closes on the predicted side

$$\alpha = \frac{1}{3} \sum_{k=1}^{3} \mathbf{1}\left[d\left(C_{t+k} - C_t\right) > 0\right]$$

$\mathcal{R}_{cc}$: return from close to close after 3 candles

$$\mathcal{R}_{cc} = d\left(\frac{C_{t+3}}{C_t} - 1\right)$$

$\mathcal{R}_{\max}$: best price reached within the 3 candles

$$\mathcal{R}_{\max} = \begin{cases} \dfrac{\max H_{t+1:t+3}}{C_t} - 1 & \text{LONG} \\ 1 - \dfrac{\min L_{t+1:t+3}}{C_t} & \text{SHORT} \end{cases}$$

$\mathcal{R}_{\min}$: worst price reached within the 3 candles

$$\mathcal{R}_{\min} = \begin{cases} \dfrac{\min L_{t+1:t+3}}{C_t} - 1 & \text{LONG} \\ 1 - \dfrac{\max H_{t+1:t+3}}{C_t} & \text{SHORT} \end{cases}$$

Main Empirical Results

4h directional accuracy (%) from the paper's Table 2; the 1h results appear only as a chart (Fig. 5)

	BTC	CL	DJI	ES	NQ	QQQ	SPX	VIX	Mean
Random	45.0	41.0	47.0	51.0	43.7	47.3	47.3	46.3	46.1
LR	46.0	54.3	52.0	43.0	48.7	56.0	59.7	48.7	51.1
XGBoost	45.3	40.0	47.3	52.0	47.3	52.7	60.0	53.3	49.7
QuantHarness	**50.7**	**55.0**	**52.3**	**55.0**	**55.3**	**59.7**	**63.7**	**54.7**	**55.8**

The paper's claims for QuantHarness:

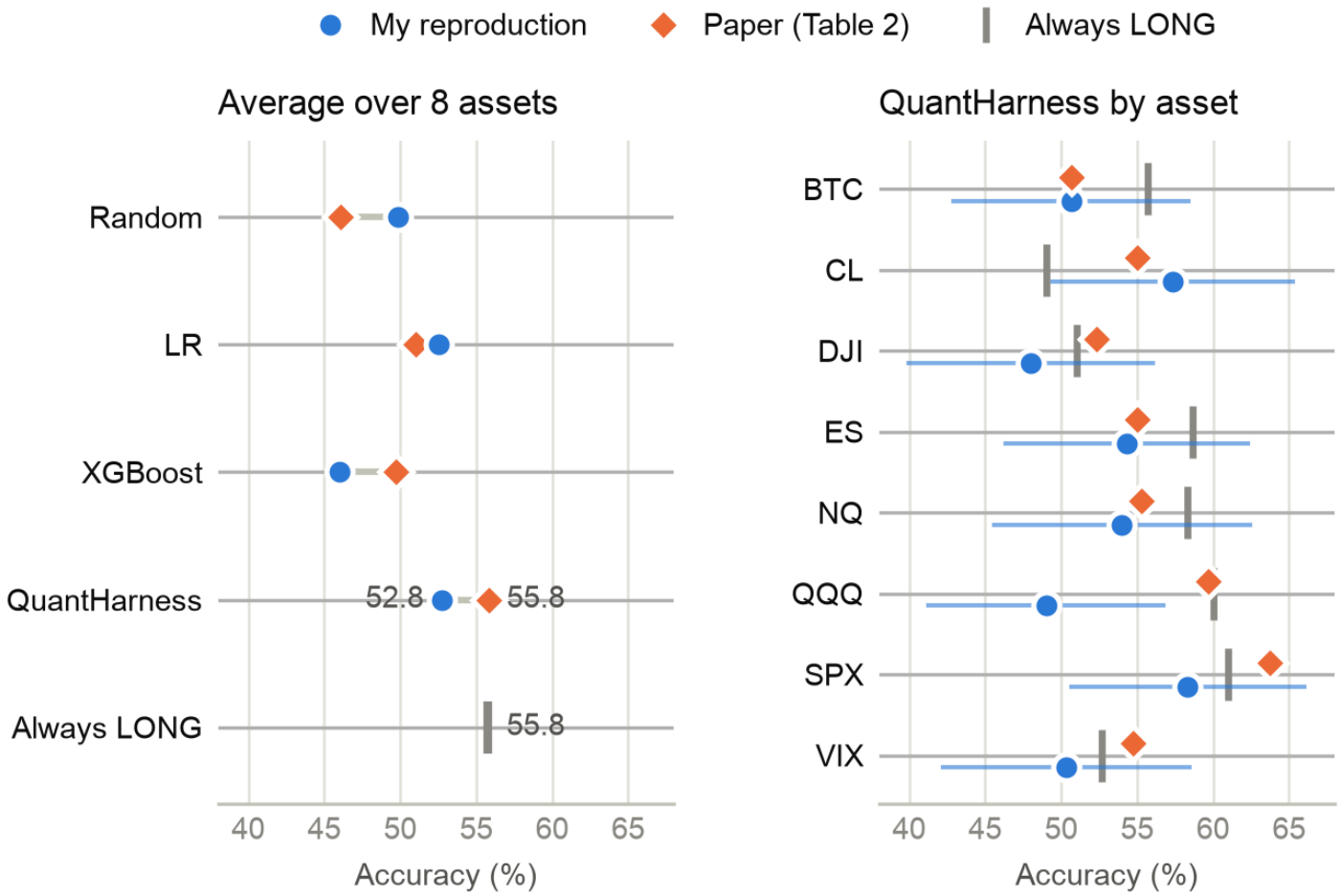
- Highest accuracy on all 8 assets
- Highest $\mathcal{R}_{cc}$ on 7 of 8
- Highest $\mathcal{R}_{\max}$ and least negative $\mathcal{R}_{\min}$ on 6 of 8 each

Experiment Reproduction

	Paper / repo	This reproduction
QuantHarness implementation	The web UI's `tail(45)` is built for live data; on a benchmark segment it would leak the 3 outcome candles	Adapted so the agents see only the last 45 of the 97 visible candles
Data	100 segments per asset	The same files from the repo
Model	Not named; the repo default is gpt-4o	Gemini 3.5 / 3.1 Flash-Lite (98% of segments)
Random	Seed not stated	Averaged over 100 seeds
LR	Last 40 closes	Same; the paper's numbers match a window 3 candles earlier
XGBoost	Random 50/50 split of files	Chronological split, so training never overlaps the test period
Returns	Version 4 describes stop-loss and take-profit caps	Uncapped (version 2), matching the reported numbers

differs from the paper

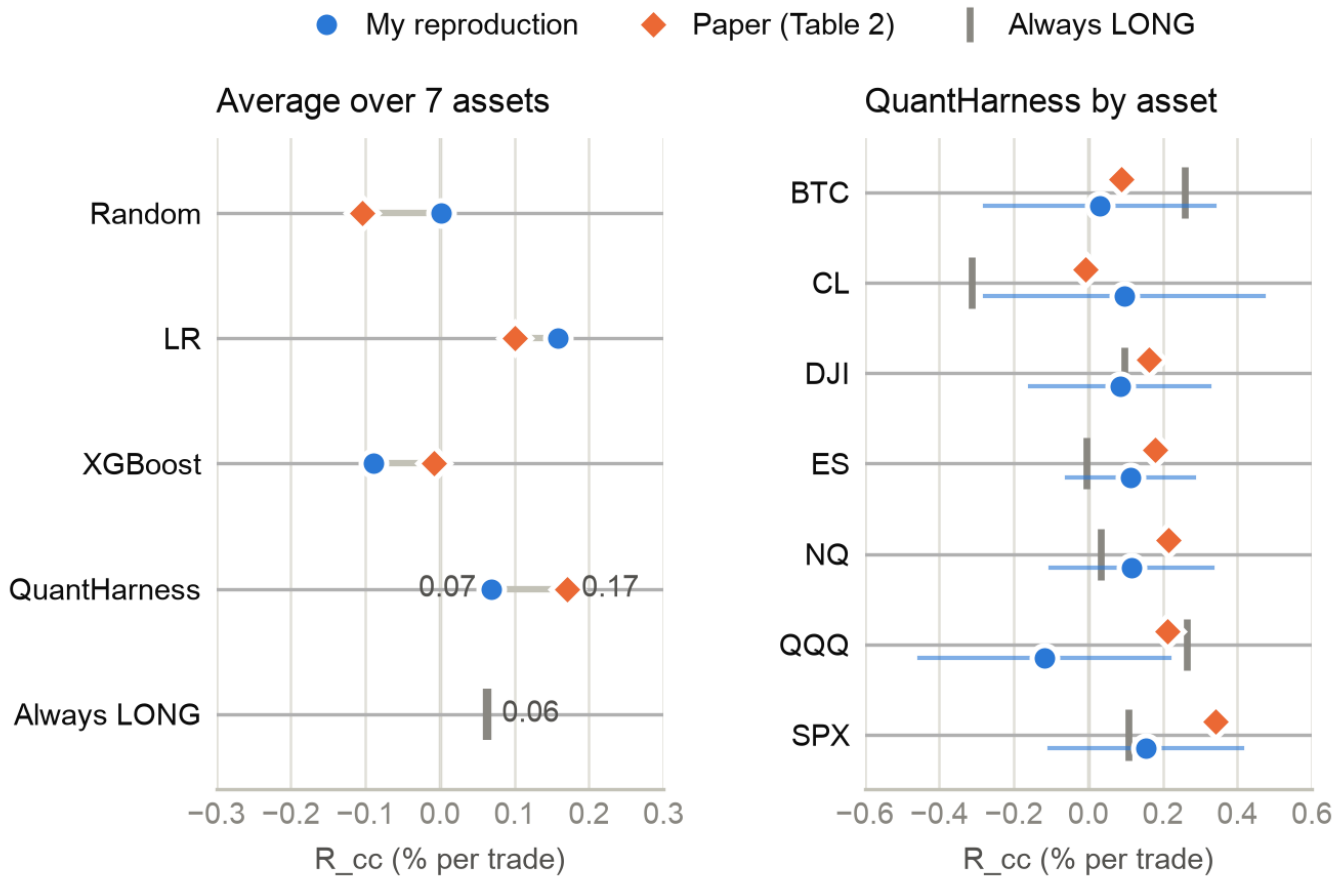
Reproduction results



4h, 100 segments per asset. Random averaged over 100 seeds, XGBoost with a chronological split, QuantHarness mostly Gemini Flash-Lite; lines are 95% intervals

QuantHarness reaches **52.8%** vs the paper's **55.8%**, within noise on 7 of 8 assets, but it is level with LR and below always-LONG (55.8%)

Reproduction results: returns



4h, uncapped close-to-close return over the next 3 candles. VIX is left out: its price ranges are 1.8 times the paper's; lines are 95% intervals

QuantHarness earns **0.07%** per trade vs the paper's **0.17%**: level with always-LONG, slightly below LR. $\mathcal{R}_{\max}$ and $\mathcal{R}_{\min}$ mostly track volatility

Strengths and Limitations

Strengths

- Uses only price data, so no lagging or noisy text sources
- Every call comes with indicator, pattern and trend reports and a justification, so the result is interpretable
- No training is required, and the code supports several LLM providers
- The code and the benchmark data are open

Limitations

Stated by the authors

- Accuracy drops on ultra-short candles ($\approx$ 1–15 min; no numbers given)
- Not exactly real-time since each decision takes several LLM calls

Found in this reproduction

- Does not noticeably outperform simple approaches: level with Linear Regression, below always-LONG
- The paper's gains are measured against an unusually low Random baseline (below 99 of 100 seeds)
- The paper and the actual code differ: no RiskAgent or stop-loss in the code, and the paper's returns are uncapped despite the caps it describes

Suggested improvements

- **Justify the use of LLMs.** Most LLM-based approaches use text data for reasoning, while QuantHarness uses only price history. Its LLM calls cost more time and resources, yet the paper does not show that language-based reasoning beats simpler and faster numerical methods.
- **Use LLMs where language helps.** LLMs can be used for formulating hypotheses, generating backtesting code, interpreting diagnostic results, etc.
- **Keep per-candle decisions numerical.** Reading RSI, MACD and candle shapes with an LLM every hour may be an expensive and unreliable way to run what could be a simple numerical rule.

The main takeaway: LLMs may be more useful for researching, designing and explaining trading strategies than for making individual trading decisions.